

Project Charter

Behind the Seens — The Autism & Emotional AI Platform

PROJECT NAME Behind the Seens — The Autism & Emotional AI Platform	DATE September 2026 (Assumed Project Kickoff)
PROJECT MANAGERS Ezz El-Din, Ibrahim, Ahmed, Mariam, Arwa, Hoor — PM Leadership Team (6 members).	PROJECT SPONSOR NTI – Digital Egypt Youth Initiative (placeholder sponsor for simulation purposes)

PURPOSE

To give parents early, accessible insight into their child's emotional state and the possible presence of Autism Spectrum Disorder (ASD) indicators by analyzing the child's drawings with AI — connecting them with qualified experts to turn that insight into a practical intervention plan, without replacing a formal medical diagnosis.

Impact: Reduce the average time between the first observable signs and a family's first real support step, especially for families with limited access to fast, specialized clinical assessment.

DESCRIPTION

An AI platform (mobile app + Expert Dashboard) that receives a child's drawing along with a short contextual questionnaire, analyzes it through a hybrid pipeline (Computer Vision + Vision-Language Model), classifies the emotional state and calculates an ASD probability score via an ensemble model, generates a simplified narrative report for parents via an LLM, and is reviewed continuously by a full-time Clinical Advisory Committee consisting of 3 psychological consultants (Human-in-the-Loop) who build an ABLLS-R-based intervention plan — followed by parents through a dedicated dashboard with a RAG chatbot for support.

OBJECTIVE

- Launch a usable MVP within 18 months, serving a pilot group of at least 50 parent–child pairs and 5 therapists/psychiatrists.
- Achieve an emotional-classification model accuracy of $\geq 80\%$ and ASD-screening reliability of $\geq 80\%$ concordance with expert clinical rating, measured on the pilot dataset by Month 16 (interim target — to be finalized with the Clinical Advisory Committee during the QA phase).
- Reduce the average time between a parent's first concern and a first expert-reviewed insight to under 2 weeks, down from the current multi-month wait for a standard clinical assessment in Egypt.
- Onboard at least 3 partner schools for pilot screening access by Month 18, as a first step toward the tool's use beyond individual households.
- Build trust among parents and experts in the tool as a first-line support layer, not a diagnostic replacement — measured via a pilot-phase satisfaction score of $\geq 4/5$ from participating parents and therapists.

PROJECT SPECIFICATION

- Support image upload and analysis with a target end-to-end response time of under 5 minutes per case (interim target, pending load testing in QA).
- Maintain children's data privacy in line with health-data protection standards.
- Mandatory human review (Human-in-the-Loop) by the Clinical Advisory Committee before any recommendation reaches parents.

BUDGET

Estimated total: EGP 4,200,000 over 18 months (simulation estimate), allocated as:

- **7-person Tech Execution Team salaries (45% = EGP 1,890,000 total):** Covers 7 tech specialists at an average of EGP 15,000/person/month.
- **3-person Clinical Advisory Committee salaries (~26% = EGP 1,080,000 total):** Covers 3 full-time psychological consultants at EGP 20,000/person/month.
- **Cloud / GPU compute for model training & inference (15% = EGP 630,000 total):** Allocated for model training, VLM/LLM hosting, and API infrastructure.
- **QA / UAT testing, pilot rollout, and miscellaneous expenses (~14% = EGP 600,000 total):** Covers QA testing, pilot school setup, legal/privacy compliance, and project contingency.

DELIVERABLES

- Parent mobile app (Data Input + Parent Dashboard)
- Expert Review Dashboard
- Full AI pipeline (CV + VLM + Ensemble Model + Fine-tuned LLM)
- RAG Chatbot for parental support
- Validation report on accuracy and reliability, signed off by the Clinical Advisory Committee
- Signed pilot agreements with at least 3 partner schools

IN SCOPE

- Drawing analysis for emotional tracking + early ASD screening
- Human expert review and an ABLLS-R-based intervention plan
- Parent dashboard with tracking, home tasks, and a chatbot
- Pilot rollout through partner schools (screening access only — not school-run assessments)

OUT OF SCOPE

- Any final or formal medical diagnosis
- Direct treatment or medical intervention
- Analysis of non-drawing inputs (video/audio) at this stage

PROJECT RISK

- **Clinical expert availability** — the project depends on the Clinical Advisory Committee at two hard schedule gates (Month 5: ASD indicator criteria; Month 16: clinical validation & UAT sign-off). Delays here stall the whole timeline, not just one workstream.
- ASD screening accuracy & ethical/legal sensitivity — these are treated as one combined risk: low accuracy directly creates ethical and legal exposure, since this is a screening tool used on children.
- Limited drawing data — insufficient volume or inconsistent quality of children's drawings could undermine both model development and validation reliability; the team's risk analysis rates this High on both probability and impact.
- Child data privacy remains a tracked risk (see Risk Management Plan) but is rated Medium severity and is not in the top 3 above.

TIMEFRAME

Start: September 2026 → End: February 2028 (18 months total)

- Month 5 — Foundation ready — data collected; ASD indicator criteria agreed with the Clinical Advisory Committee.
- Month 10 — First AI version — emotional + ASD scoring models built, ready for internal testing.
- Month 13 — Everything connected — mobile app and dashboards fully linked to the AI pipeline.
- Month 16 — Clinical sign-off — Clinical Advisory Committee validates accuracy; User Acceptance Testing (UAT) complete.
- Month 18 — Launch — MVP goes live; project handed over.

PROJECT MANAGER AUTHORITY

Strategic and cross-artifact decisions (e.g. scope changes, budget reallocation, schedule changes affecting milestone gates) are made by majority vote among the 6 PM Leadership Team members. No single member holds unilateral decision-making authority; each member drafts and maintains their assigned artifact (Charter, Schedule, Resource Allocation, Risk Management, Communication Plan, QA Plan), with the full team reviewing and voting before an artifact is finalized.

KEY STAKEHOLDERS

Stakeholder	Role / Interest
Parents & children (end users)	Rely on the accuracy, clarity, and sensitivity of the reports provided.
Clinical Advisory Committee (3 Full-time Psychological Consultants)	Review the system's reports continuously behind the AI and build intervention plans on top of them; their sign-off is required before any result reaches a parent.
Execution team (7 tech specialists)	Responsible for building and delivering every technical component of the platform through a lean, cross-functional engineering approach.
Program Sponsor	Provides funding and tracks progress against agreed objectives.
Child data-protection / ethical compliance bodies	Ensure the platform meets privacy and medical-ethics standards for minors.

TEAM ROLES & RESPONSIBILITIES

PM / Leadership Team (6 members — Ezz El-Din, Ibrahim, Ahmed, Mariam, Arwa, Hoor): oversees planning, tracking, risk management, and cross-stakeholder communication — not directly involved in technical execution.

Execution Team (10 specialists: 7 tech + 3 clinical, reporting to the PM team):

Role	Responsibility
Data Engineer	Collects, cleans, and stores drawing data.
AI Engineers (3 specialists)	Collects and cleans drawing data, builds CV & ensemble models for ASD screening, fine-tunes VLMs/LLMs for report generation, and manages MLOps.
Full-stack Developer	Connects the pipeline to applications via APIs and builds the Expert Review Dashboard interface, working with the UX/UI Designer.
Mobile Developer	Builds the parent app (Data Input + Parent Dashboard).
UX/UI Designer	Designs a sensitive experience for parents and experts.
QA Engineer	Tests accuracy and reliability, and coordinates UAT.
Clinical Advisory Committee (3 Full-time)	Forms a continuous consulting board to review AI outputs, reviews ABLLS-R criteria, and validates result accuracy.

Ethical Constraint: The system is a supportive screening / analytical tool — NOT a replacement for medical diagnosis.